Teach the complete learning journey

Purpose, readiness, pacing, projects and evidence of understanding.

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What this programme is for

Children learn to understand, design, build, test and explain AI systems responsibly. The same topic is taught through a complete student lesson, a matching workbook and a teacher guide with answer reasoning. Each level contains 36 weeks. Typical placement runs from Grade 6 to Grade 12; demonstrated readiness matters more than age alone.

The first four levels preserve the original Curriculum 2.0 sequence and projects. Later explanations are integrated into the required lessons. Three additional levels cover neural networks, AI engineering and research. Twelve Python bridge lessons prepare learners for the coding pathway. All 144 original weeks and all 252 later source lessons are represented.

Start with the learner, then choose the level

Ask the learner to explain an unfamiliar case before supplying vocabulary. Use each week's readiness question to check the actual prerequisite. A memorised definition does not demonstrate that a learner can trace a procedure, calculate a result or distinguish supplied facts from guesses.

- Foundations: observe, count, compare and explain a simple rule.
- Algorithms and Data: trace branches and loops; explain the difference between a label and a feature.
- Generative AI: explain context, source evidence, fractions and responsible use of approved material.
- Machine Learning: complete the Python bridge or demonstrate equivalent skills; use coordinates, fractions and functions.
- Neural Networks: use signed numbers, linear equations, coordinates and Python traces.
- AI Engineering: understand JSON, permissions, model evaluation and grounded generation.
- Research: control variables, preserve experiment records, compare baselines and explain uncertainty.

Use the Python bridge before the Machine Learning coding lessons. Its twelve lessons are additional preparation, not twelve of the 252 numbered weeks. File handling and exception recovery are taught in the integrated Machine Learning readiness week.

Plan the time honestly

Start with three 45-minute sessions per week for the core explanation, guided practice and independent case. The named connected topics add teaching time: reserve further sessions rather than trying to rush every example into that starting allocation. Investigations and project construction may need separate laboratory time. Teacher timing tables describe individual examples; do not add overlapping tables as though they describe one fixed-duration lesson.

1. Retrieve the prerequisite and record a prediction.
2. Model the mechanism with the worked trace and visual.
3. Compare a changed input, diagnose a mistake and practise together.
4. Run or trace the investigation, preserve actual evidence and connect the project.
5. Attempt the independent case; give feedback and reassess with a genuinely changed case.

Use the headings as a connected teaching route. All named connected examples are required content; split them into additional sessions when necessary. A supported attempt is practice evidence. Mark it as supported and collect a fresh independent attempt later.

Teach with the full weekly model

Each student week retains fourteen original connected sections, plus a lesson route and finish-and-return check: mission, observation, mechanism, comparison, vocabulary, worked example, error diagnosis, guided practice, investigation, mastery, project connection, limits, complete case and independent transfer. Two original Machine Learning weeks also retain a fifteenth executable laboratory.

The teacher guide retains preparation, explanation, guided teaching, misconceptions, assessment answers and project progression. The original two coding weeks retain their additional laboratory guidance. Workbooks keep the original activities and add the integrated examples. Teacher answers belong in teacher copies; distribute the student bundle to learners.

Read the accompanying explanation as well as the diagram. Short figure labels are intentionally abbreviated. Ask students to explain the intermediate steps, the source of each fact and the limit of the conclusion. Use the interactive Foundations Week 3 light slider at 29 and 30 to test the exact boundary.

Build seven cumulative projects

Helpful Classroom Sorter

A paper or block-based sorting prototype with an explicit input policy, written rule, unsupported-input response and reviewed fictional cases.

Evidence to defend: Compare against a simple fixed sorting rule. Test ambiguous cards, missing features and a changed user need; explain what would require learning rather than a written rule.

Transparent Book Recommender

A recommendation procedure using approved fictional readers and books, declared features and visible score calculations.

Evidence to defend: Compare with a simple popularity or fixed-rule recommendation. Test different readers and feedback effects; explain who is represented and why personal information is unnecessary.

Verified Study Buddy

A grounded assistant design or mock prototype that distinguishes supplied source evidence from unsupported generated claims.

Evidence to defend: Compare with a source-only answer procedure. Record retrieval evidence, output checks, citations, abstentions, unsafe requests and at least one case where generation should not proceed.

Responsible Plant Classifier

A reproducible small classifier investigation with dataset and model cards, split records, baseline and error analysis.

Evidence to defend: Keep training, validation and final evaluation separate. Report counts and task-relevant metrics, subgroup limitations and a human-review policy; a toy success does not establish reliable plant diagnosis.

Image-pattern Network Investigation

A small network investigation using fictional or approved image patterns, traced layer calculations and versioned training/evaluation evidence.

Evidence to defend: Compare a simple baseline. Change one architecture or training choice at a time. Preserve losses, per-case results, seeds or deterministic inputs and an ablation; explain what each result fails to establish.

Grounded Learning Service

A mock or local service combining approved retrieval, structured requests, role permissions, validation and operation records.

Evidence to defend: Compare retrieval and answer baselines. Test citation support, injection boundaries, missing context, access rules, repeated operations, failures, latency and resource limits. Use a mock tool ledger for side effects.

Reproducible AI Research Study

An investigation with a bounded claim, prior work, comparison conditions, repeatable procedure and preserved artefacts.

Evidence to defend: Control relevant variables and distinguish observation from causal claims. Report repeated runs where meaningful, sample scope, exclusions, uncertainty and unsuccessful results. Another learner must be able to reproduce the procedure.

Keep a cumulative evidence portfolio

Use one portfolio per level, with a separate entry for each week. Keep the input, rule or model version, prediction, actual result or “not run”, first mismatch, repair, limitation and support used. Reference previous entries when a new concept changes a component. Do not silently replace unsuccessful results.

Weeks	Portfolio milestone
1–6	Explain the task and mechanism; retain observation and readiness evidence.
7–12	Define permitted data, assumptions, a simple comparison and failure cases.
13–18	Trace or construct components; preserve intermediate steps and first errors.
19–24	Connect components and examine changed inputs, source quality and held-out evidence.
25–30	Compare versions under the same declared conditions; document safety and resource limits.
31–32	Freeze the project claim, success criteria, evaluation procedure and review responsibilities.
33–35	Build, evaluate and revise the final project using the level’s named capstone lessons.
36	Defend the evidence, limits and reproducibility; complete a fresh transfer case.

These are portfolio milestones, not replacements for the specific weekly instructions. Define data splits before tuning any trained model. Revisit a component only when the relevant concept and prerequisite have been taught.

Assess understanding and independence

Dimension	Beginning	Developing	Independent
Mechanism	Names a tool or repeats a definition.	Traces part of the procedure with prompts.	Explains every relevant step on a changed case.
Evidence	Treats a guess or expected output as a measurement.	Records results but needs help checking the comparison.	Separates prediction and observation; preserves inputs, versions and fair comparisons.
Limits and responsibility	Claims general success from one example.	Names a risk without a concrete response.	Links a specific failure to a bounded claim, fallback or human review.

Transfer	Copies the supplied answer.	Solves a similar case with support.	Solves and explains a fresh case without hints.

Record each dimension separately; a polished presentation must not conceal weak reasoning. Use the matching teacher answer to judge the supplied case, not as the only acceptable wording. If an alternative procedure meets the stated contract, ask for a complete trace and fair comparison.

To finish a level, require an independent changed case, a defensible project evidence pack and an explanation of one important failure. An unresolved prerequisite needs targeted practice and fresh reassessment before the dependent level.

Make the lesson accessible

Preteach a few terms, read one paragraph at a time and ask the learner to point to its corresponding visual step. Offer a smaller fictional case, manipulatives, a partially completed trace or oral explanation before returning to the full task. Keep the same reasoning goal. For extension, change one assumption, add a boundary case or challenge the evidence claim rather than merely adding more questions.

Allow a spoken or drawn explanation when writing is the barrier. Record any support so it is clear whether the result demonstrates independence. The HTML works on phones; tables scroll within their panels. The Word copies are editable, and PDF copies provide printable response space.

Prepare tools and distribute the right files

The HTML lessons, images and navigation work offline. Open START_HERE.html directly in a browser. No deployment, account, paid API or mounted disk image is required for the supplied teaching package. The example Python programs use the standard library and fictional local inputs; use a teacher-approved Python environment when running them. Paper tracing is useful preparation, but mark execution as “not run” until it actually happens.

Use the student bundle for learners: it includes student lessons, workbooks, images and their downloads. Keep teacher files, canonical JSON and source inputs in the teacher collection because they contain solutions. The complete folder is a private teaching package, not an access-controlled website. Draft answers stay in the current browser when storage is available. Print or save a PDF for your portfolio, and clear drafts on shared devices.

Corrections and review status

This edition repairs specific inherited problems: an unrelated worked explanation in Foundations Week 4; a distance-versus-destination transfer task; an overconfident label-agreement inference; an overgeneralised image-generation description; impossible confusion-matrix rates; an unnamed distance metric; ambiguous probability semantics; and an error-audit overlap counted as an extra failure. Seven inconsistent worked-image cards are replaced with matching native diagrams; original cards remain only in private source

provenance.

The content audit checks all original blocks, the integration of every later source lesson, question/teacher-answer alignment and image availability. Desktop and phone browser checks cover navigation, pictures, drafts, progress, readiness and printing. Current results are in reports/classroom-browser-validation.json. PDFs are generated with the local PyMuPDF renderer. Code examples are run, including the deliberately failing indexing demonstration. Download checks compare actual Word and PDF text with their matching canonical role content. These are editorial and technical checks; classroom pacing and age suitability still need a teacher's judgement and observation of real learners.

Topic-specific pictures and weekly end goals

Every main week and Python readiness lesson now includes an agent-reviewed worked case, a labelled pictorial trace, a concrete explanation question, a teacher key and project evidence. A further 110 numerical pictures show actual matrices, plotted coordinates, weighted calculations and comparisons. Read these alongside the supplied facts and explanations. The separately named independent case remains reserved for an attempt before feedback.

Explore the pictorial lesson index. The complete original curriculum is preserved; these worked cases add teaching support.

The unified sequence

Level 1 Foundations → Level 2 Algorithms and Data → Level 3 Generative AI Systems → Python Readiness Bridge → Level 4 Machine Learning → Level 5 Neural Networks → Level 6 AI Engineering → Level 7 Research and Advanced Projects.

The bridge is preparation, without a level number. Use the hosted readiness check, then confirm skills on another independently written and tested program before skipping it.

Find the lesson route and keep your evidence

Each hosted lesson opens with six links: wonder and remember, watch and explain, try with help, investigate and repair, try independently, and keep project evidence. The complete original sections remain available. Each week adds a teacher preparation summary, a specific question about the reviewed picture, and a project portfolio milestone.

Hosted drafts and progress stay in the current browser. Download or print evidence for a lasting portfolio; clear drafts on a shared device. A progress tick records practice, not demonstrated mastery. Teachers judge explanation, trace or test, and limits as not yet, with support, or independent.

Practice and keep evidence

Use supported questions, hints and changed cases to practise a worked explanation. Record completed answers and reflections, then download them using the portfolio controls. Practice feedback does not replace independent assessment.

Supported Python examples run in a browser worker with real output and a five-second execution limit. Predict first, run, inspect errors, change one thing and test again. Earlier lesson links provide suggested repair tasks.

Offline Python needs a local HTTP server: run `python3 -m http.server 8000 --bind 127.0.0.1` from the programme folder, open the local student index, and stop the server after use. Reading and paper tracing remain available when opening files directly.

Run a real classroom pilot

Use the teacher pilot studio on the website or the offline pilot guide. Three readiness tracks offer baseline, independent post-lesson and delayed tasks. Record real anonymous learner evidence, support, exposure, time and barriers before revising the lesson. No real classroom outcomes have yet been collected.